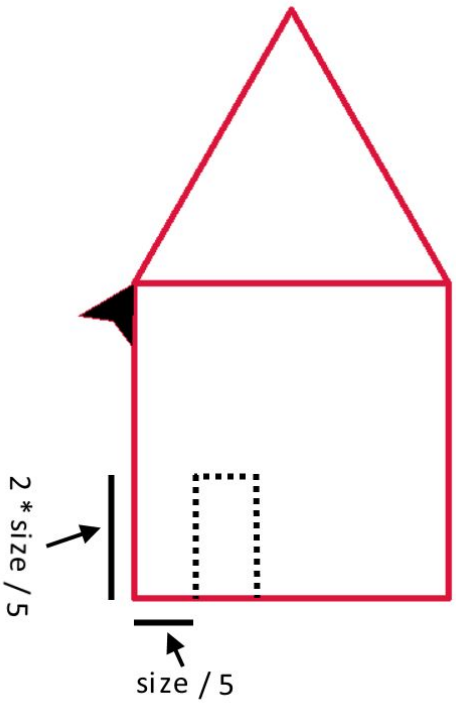
ISTA 130 Math Worksheet

Given the partially drawn house to the right with the following rectangle function, finish the incomplete house function below so that calling it will draw a door in the position shown by the dotted lines with the specified relative dimensions. Then rewrite `rectangle` using a `for` loop.




```
def rectangle(torty, width, height):
    torty.forward(height)
    torty.right(90)
    torty.forward(width)
    torty.right(90)
    torty.forward(height)
    torty.right(90)
    torty.forward(width)
    torty.right(90)

def house(torty, size, x, y):
    torty.penup()
    torty.goto(x, y)
    torty.pendown()
    torty.setheading(90)
    square(torty, size)
    torty.forward(size)
    torty.right(30)
    triangle(torty, size)
```

(Continued on other side)

Write a function called `average` that takes three numbers as arguments and prints their average.

The formula for the volume of a cylinder is $\pi * r^2 * h$, where r is the radius of the circular base and h is the height. Write a function called `cylinder_V` that takes the radius and height as arguments and prints the volume of the cylinder. For π, write `math.pi`.

```
import math
```

Recall from lecture that the formula for the turn angle for a turtle to draw a polygon with `n` sides is: `angle = 360 / n`. Write a function called `octadecagon` that takes a turtle and a side length as arguments and draws a 18-sided polygon with sides of the given length. Use a `for` loop.

Write a function called `polygon` that takes a turtle and a number of sides and a side length as arguments and draws a polygon with the specified number of sides of the given length.